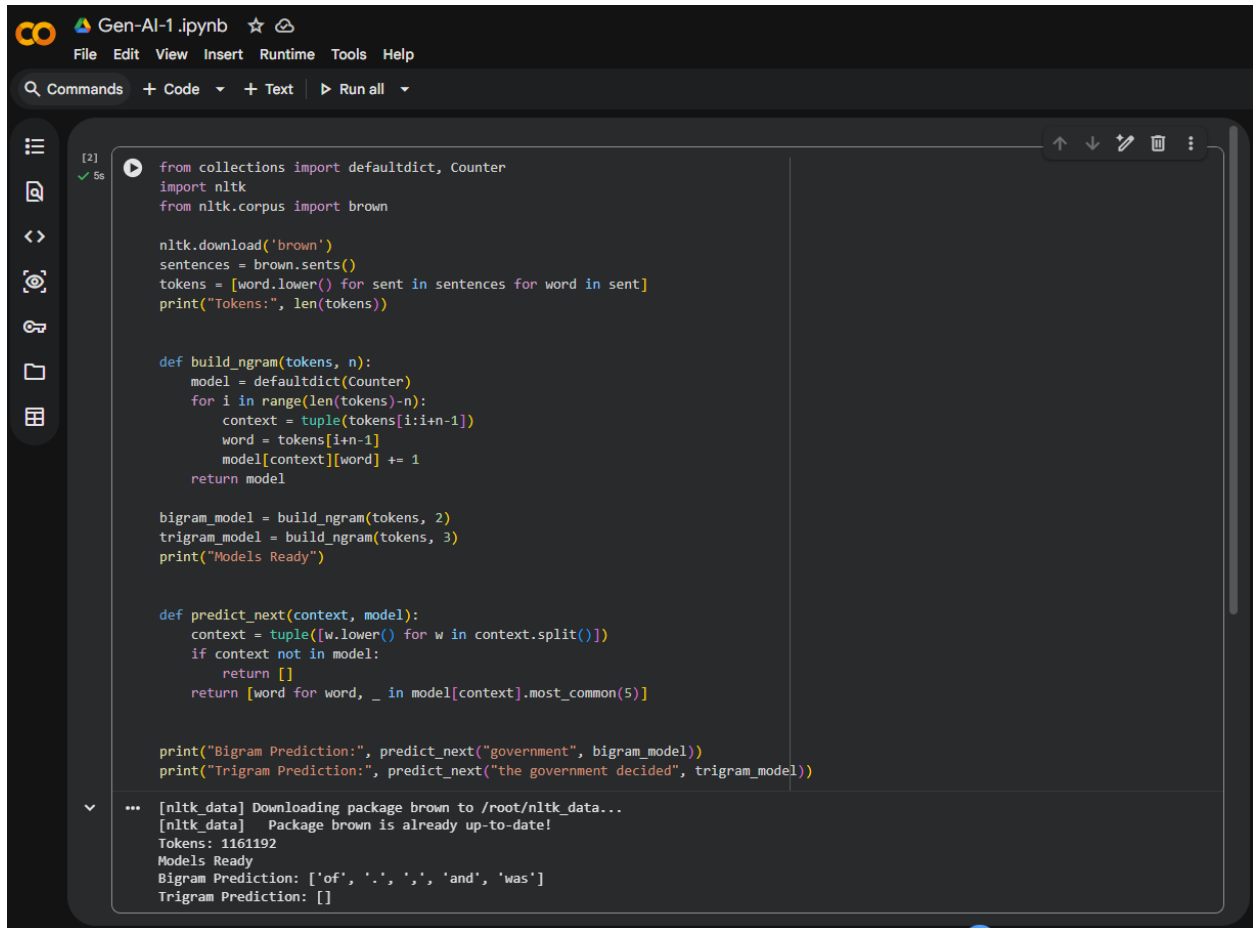


EXP - 1 A LAB EXPERIMENT



The screenshot shows a Jupyter Notebook titled "Gen-AI-1.ipynb" with a dark theme. The interface includes a top menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a command palette (Commands), and a left sidebar with icons for file explorer, search, and other tools. The main area contains a code cell with the following Python code:

```
[2] ✓ 5s
from collections import defaultdict, Counter
import nltk
from nltk.corpus import brown

nltk.download('brown')
sentences = brown.sents()
tokens = [word.lower() for sent in sentences for word in sent]
print("Tokens:", len(tokens))

def build_ngram(tokens, n):
    model = defaultdict(Counter)
    for i in range(len(tokens)-n):
        context = tuple(tokens[i:i+n-1])
        word = tokens[i+n-1]
        model[context][word] += 1
    return model

bigram_model = build_ngram(tokens, 2)
trigram_model = build_ngram(tokens, 3)
print("Models Ready")

def predict_next(context, model):
    context = tuple([w.lower() for w in context.split()])
    if context not in model:
        return []
    return [word for word, _ in model[context].most_common(5)]

print("Bigram Prediction:", predict_next("government", bigram_model))
print("Trigram Prediction:", predict_next("the government decided", trigram_model))
```

Below the code cell, the output is displayed:

```
... [nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Package brown is already up-to-date!
Tokens: 1161192
Models Ready
Bigram Prediction: ['of', '.', ',', 'and', 'was']
Trigram Prediction: []
```